Vision of the Department

To be a well-known centre for pursuing computer education through innovative pedagogy, value-based education and industry collaboration.

Mission of the Department

To establish learning ambience for ushering in computer engineering professionals in core and multidisciplinary area by developing Problem-solving skills through emerging technologies.

Session 2025-2026

Vision: Dream of where you want.	**Mission:** Means to achieve Vision

Program Educational Objectives of the program (PEO): (broad statements that describe the professional and career accomplishments)

PEO1	**Preparation**	**P: Preparation**	**Pep-CL abbreviation pronounce as Pep-si-lL easy to recall**
PEO2	**Core Competence**	**E: Environment (Learning Environment)**	
PEO3	**Breadth**	**P: Professionalism**	
PEO4	**Professionalism**	**C: Core Competence**	
PEO5	**Learning Environment**	**L: Breadth (Learning in diverse areas)**	

Program Outcomes (PO): (statements that describe what a student should be able to do and know by the end of a program)

Keywords of POs:
Engineering knowledge, Problem analysis, Design/development of solutions, Conduct Investigations of Complex Problems, Engineering Tool Usage, The Engineer and The World, Ethics, Individual and Collaborative Team work, Communication, Project Management and Finance, Life-Long Learning

PSO Keywords: Cutting edge technologies, Research

"I am an engineer, and I know how to apply engineering knowledge to investigate, analyse and design solutions to complex problems using tools for entire world following all ethics in a collaborative way with proper management skills throughout my life." *to contribute to the development of cutting-edge technologies and Research*.

Integrity: I will adhere to the Laboratory Code of Conduct and ethics in its entirety.

Name and Signature of Student and Date
(Signature and Date in Handwritten)

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Department of Computer Technology

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Session	**2025-26 (ODD)**	**Course Name**	**Operating System Lab**
Semester	**5**	**Course Code**	**23IOT1504**
Roll No	**64**	**Name of Student**	**Sheikh Wasimuddin**

Practical Number	**03**
Course Outcome	1. Understand Computer System Configuration and Simulate system resources efficiently using Linux Commands (CO1) 2. Analyse operating system functionalities utilizing system calls, thread programming and process scheduling algorithms (CO2) 3. Apply Synchronization primitives to implement a Deadlock-free solution(CO3) 4. Simulate Disk scheduling, Memory allocation, File allocation, page replacement algorithms (CO4)
Aim	Implement thread life cycle: Thread creation & Termination. • File Downloader Simulation Create a thread that simulates downloading a file in chunks (1–100%). Use sleep() to slow down the process, and terminate after download completion.
Theory	A thread is the smallest unit of a process that can be scheduled by the CPU. In Java (or other languages), threads pass through different states, which form the thread life cycle: **1. New (Created):** o When a thread object is created using the Thread class or implementing Runnable. o Example: Thread t = new Thread(); o At this stage, the thread is not yet running. **2. Runnable:** o After calling start(), the thread enters the runnable state. o It is ready to run but waiting for CPU scheduling. **3. Running:** o When the CPU scheduler picks a thread, it starts executing inside the run() method. **4. Waiting / Timed Waiting (Optional):** o A thread can pause execution using methods like sleep() or join(). o In this project, we use sleep() to simulate the download delay. **5. Terminated (Dead):** o After the run() method finishes, the thread completes execution and enters the terminated state.

	○ Once terminated, it cannot be restarted.

Procedure and Execution	//sheikh wasimuddin 23071044 CSEIOT 64 OS PRACTICAL NO.3
	
	#include <stdio.h>
	
	#include <pthread.h>
	
	#include <unistd.h>
	
	void* download(void* arg) {
	  for (int i = 1; i <= 100; i++) {
	  printf("Downloading: %d%%\n", i);
	  usleep(50000);
	  }
	  return NULL;
	}
	
	int main() {
	  pthread_t t;
	  pthread_create(&t, NULL, download, NULL);
	  pthread_join(t, NULL);
	  return 0;
	}

Output:

main.c

```c
//sheikh wasimuddin 23071044 CSEIOT 64 OS PRACTICAL NO.3
#include <stdio.h>
#include <pthread.h>
#include <unistd.h>

void* download(void* arg) {
    for (int i = 1; i <= 100; i++) {
        printf("Downloading: %d%%\n", i);
        usleep(50000);
    }
    return NULL;
}

int main() {
    pthread_t t;
    pthread_create(&t, NULL, download, NULL);
    pthread_join(t, NULL);
    return 0;
}
```

```
Downloading: 1%
Downloading: 2%
Downloading: 3%
Downloading: 4%
Downloading: 5%
Downloading: 6%
Downloading: 7%
Downloading: 8%
Downloading: 9%
Downloading: 10%
Downloading: 11%
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Downloading: 28%
Downloading: 29%
Downloading: 30%
Downloading: 31%
```

main.c

```c
//sheikh wasimuddin 23071044 CSEIOT 64 OS PRACTICAL NO.3
#include <stdio.h>
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    }
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}

int main() {
    pthread_t t;
    pthread_create(&t, NULL, download, NULL);
    pthread_join(t, NULL);
    return 0;
}
```

```
Downloading: 74%
Downloading: 75%
Downloading: 76%
Downloading: 77%
Downloading: 78%
Downloading: 79%
Downloading: 80%
Downloading: 81%
Downloading: 82%
Downloading: 83%
Downloading: 84%
Downloading: 85%
Downloading: 86%
Downloading: 87%
Downloading: 88%
Downloading: 89%
Downloading: 90%
Downloading: 91%
Downloading: 92%
Downloading: 93%
Downloading: 94%
Downloading: 95%
Downloading: 96%
Downloading: 97%
Downloading: 98%
Downloading: 99%
Downloading: 100%

...Program finished
Press ENTER to exit
```

Output Analysis	When the program runs, a new thread is created using pthread_create, which immediately begins executing the download() function. The output is shown as percentages from 1% up to 100%, with a short delay between each step introduced by usleep(50000) to simulate network latency. This makes the output appear gradually, like a real file download in progress. Once the loop reaches 100%, the thread naturally completes its task and enters the terminated state. Meanwhile, the pthread_join in the main function ensures that the main program waits for the thread to finish before exiting. Overall, the output demonstrates the full thread life cycle—thread creation, execution with simulated progress, and termination—through a clear and realistic file downloader simulation.
Conclusion	In this practical, we implemented a file downloader simulation using threads in C. The program clearly demonstrated the thread life cycle where a thread is created, executes the download task in steps with simulated delays, and terminates after completion. By using pthread_create and pthread_join, we observed how threads can run concurrently with the main program and how synchronization ensures proper termination. This experiment helped in understanding thread creation, execution, and termination in a real-time scenario.
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